GROUP WORK PROJECT # 1
GROUP NUMBER: 12124

MScFE 642: Deep Learning for Finance

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

Introduction

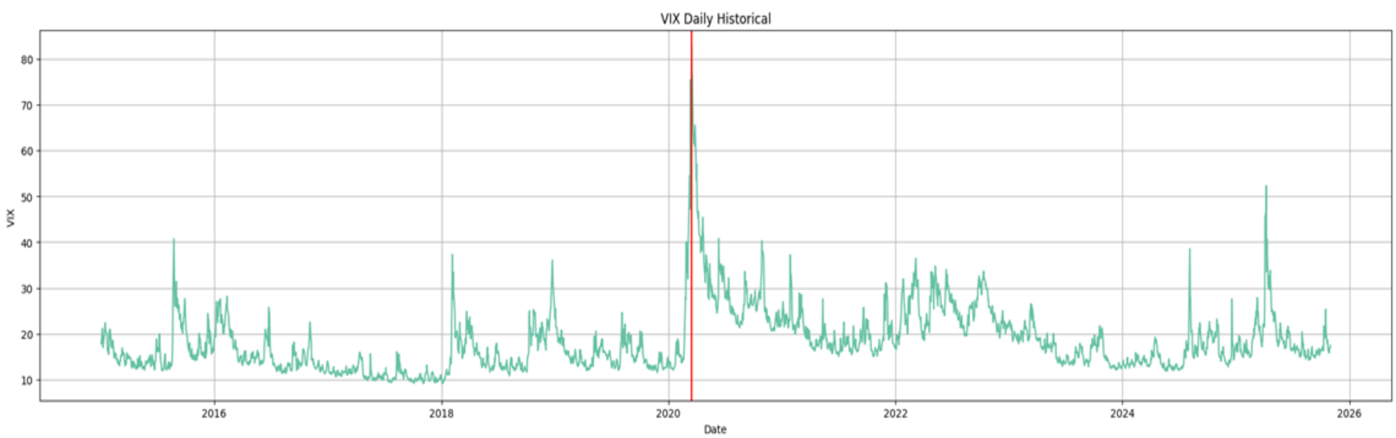
Predicting Financial Time series data can be challenging because of their nonlinearity, dynamics, and chaotic characteristics. Recently, Artificial Neural Networks have shown their power in handling the nonlinearity of this type of data(Xu et al.) In this project, we use multi-layer perceptron (MLP) and convolutional neural networks (CNN) to predict the VIX index level.

The VIX measures the implied volatility of the S&P 500 and is obtained from bid/ask quotes of S&P 500 index options with at least 8 days until expiration. It was introduced by the Chicago Board Options Exchange (CBOE) to measure the market's expectation of short-term volatility and is referred to as the "fear index".(VIX FAQ)

Why choose VIX? The VIX is a forward-looking measure; it reflects the market's expectation of future volatility over the next 30 days. This makes it different from historical volatility, which is backward-looking. Investors monitor this index to evaluate market risk, fear, and stress levels. VIX is not tradable, but by predicting it, we can profit from the potential arbitrage opportunities that arise from discrepancies between the index and its corresponding options and futures contracts (Noferesti, Hajizadeh, and Mehtari Taheri).

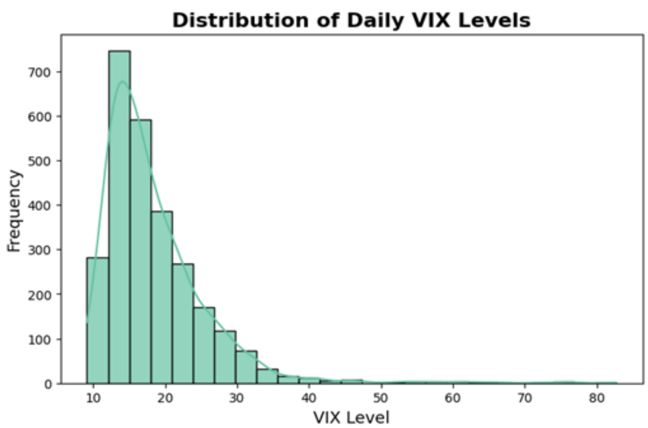
Step One

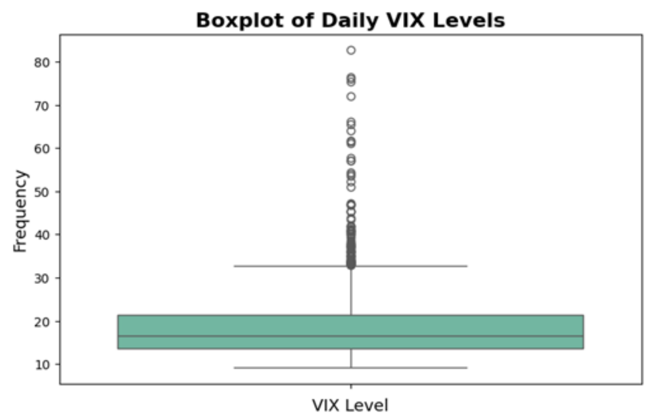
For this project, daily closing levels of the VIX were downloaded from Yahoo Finance for the period 2015-01-01 to 2025-11-01. The time series is shown below. A peak was seen in March 2020, when the investors' fear increased during the COVID-19 pandemic.



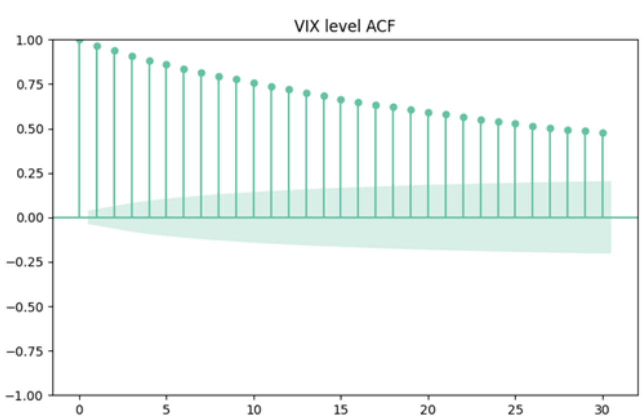
The summary statistics are shown in the table below. Over the study period, we have 2,725 observations, with a mean of 18.34 and a standard deviation of 7.17. The histogram and boxplot indicate that the distribution is right-skewed, with a median of 16.50. A VIX level between 15 and 25 is generally considered a moderate-volatility environment, while levels between 25 and 30 indicate high market volatility(Hortúa and Mora). As shown in the boxplot, VIX values above 35 are identified as outliers.

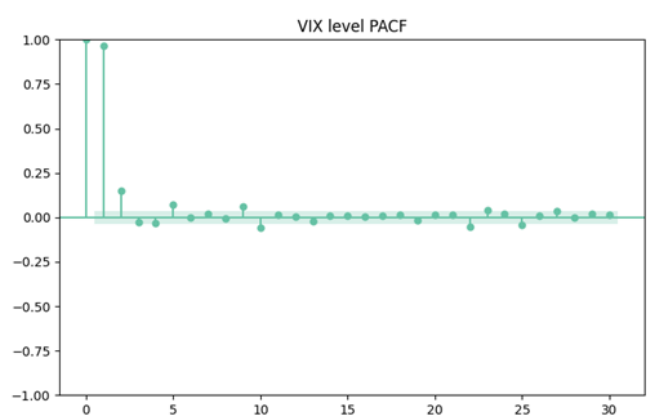
Ticker	^VIX
count	2725.00000
mean	18.342818
std	7.172723
min	9.140000
25%	13.530000
50%	16.500000
75%	21.290001
max	82.690002



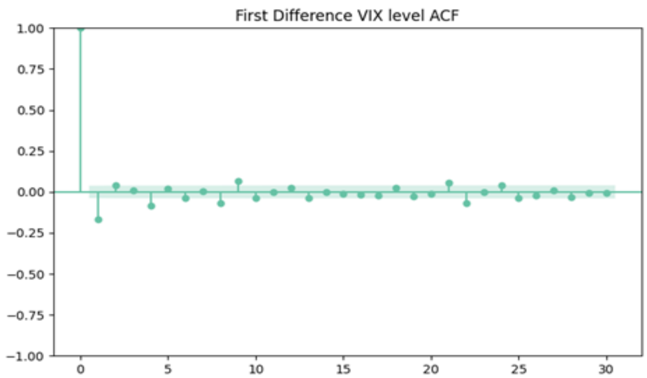


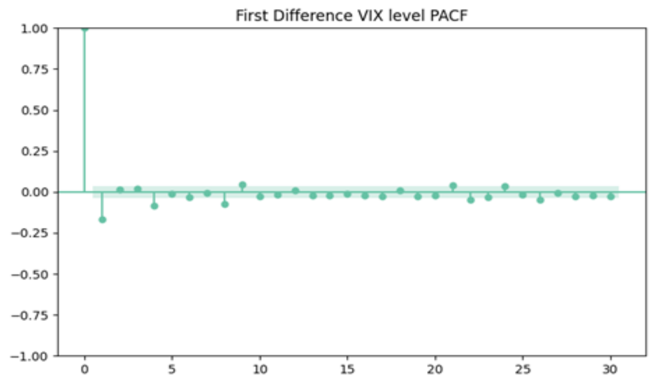
Weak stationarity means that the mean and autocovariance of a time series do not depend on time. The ACF of the VIX series shows a slow, persistent decline, indicating strong autocorrelation and the presence of a trend. This suggests that the VIX series is not stationary in levels. The PACF plot shows that the autoregressive structure is AR(1) or at most AR(2), and the PCAF dies after lag 2. So the useful prediction information comes from lag 1 and 2. However, we choose lag 1 to lag 5 to capture the weekly momentum effect and short-term persistence in volatility.





To achieve stationarity, we apply first differencing. The KPSS test (where the null hypothesis is stationarity) gives a p-value > 0.05, which means we fail to reject stationarity. The ADF test (where the null hypothesis is unit root) shows a smaller(more negative) than the 5% critical value, so we reject the null of a unit root. Together, these results confirm that the first-differenced VIX series is stationary**.**





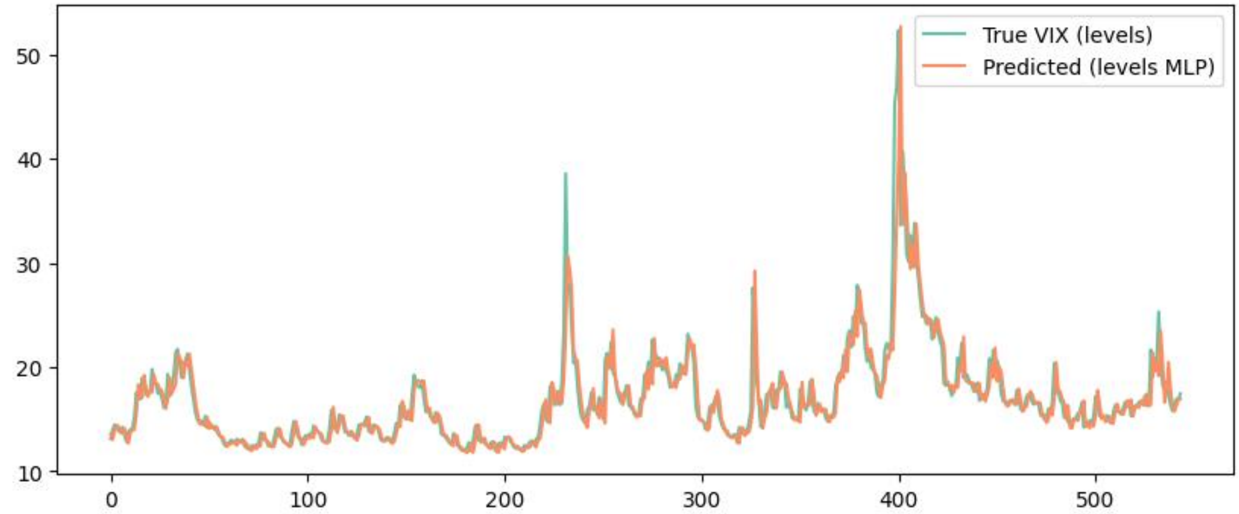
A time series becomes non-stationary when it contains memory, which means that its current values depend on its long past (e.g., long history, trends, or persistence). This memory gives predictive power to the models. However, applying first differencing can make the series stationary at the cost of removing most of this memory, which may eliminate useful predictive structure. The solution is using fractional differencing by finding the smallest fractional differencing order d that makes the time series stationary while preserving as much memory as possible from the original series(Lopez de Prado 89–112). The optimal d we find for VIX level is 0.12.

Step 2

2.1 Model 1 — MLP Using VIX Levels

In this section, we apply an MLP model to the VIX time series. As shown in the figure below, the model converges smoothly, with a stable and consistent validation loss. The predicted series closely tracks the true VIX levels and successfully captures key dynamics of the index, including:

- the gradual, low-volatility periods,
- the medium-scale oscillations, and
- the major volatility spikes (with only slight underestimation at extreme values),



2.2 Model 2 — MLP Using Stationary Log-Differences

As shown in the table below, the time series with first differencing achieves the best performance, with the lowest error among all three time series. However, this model is not without limitations: it tends to produce less volatile prediction errors, and it fails to capture large VIX jumps.

Quantitative Model Evaluation

Model	Input of the model	MSE	MAE	Interpretation
2.a	level	4.579062	1.136809	Tracks VIX well but underestimates extreme spikes
2.b	log-diff	2.476968	0.776682	More stable and accurate, but in small movements
2.c	Frac diff	70.650949	7.902078	Best trade-off

Discussion on Model performance

- The first one struggles with **high volatility jumps**, but it is **very intuitive** and apparently smooth prediction
- The second model needs predictions to be reconstructed and then **loses the ability to interpret** accurately; however, with this model, **the numerical error is improved**
- The last one balances the performance of the past two models

Step 3

In this section, we transform time series into images using the Gramian Angular Fields (GAF) encoding method. In this technique, the data is first scaled into the range [−1,1], and a series of 15 days of VIX levels is mapped into polar coordinates, where each value is represented as an angle and each timestep as a radius. A matrix is then constructed to capture temporal correlations, and images are produced that show recurring trends and anomalies visually. A CNN is then applied as it can recognize sequence patterns. CNNs can filter out noise from input data and extract valuable features or patterns. However, CNNs have a limitation in handling short-term and long-term dependencies; they cannot effectively store relevant past information to inform future predictions. This can pose a challenge when predicting the VIX, which has shown mean-reverting behavior(Salim and Djunaidy).

In the table below, we report the mean absolute error (MAE), as outliers were not removed during data processing. Although MSE values are provided in the Jupyter notebook, they can be heavily influenced by outliers.

Model	Train Error-MAE	Test Error-MAE
Naive model-VIX Level	18.630264	17.166730
GAF-CNN-VIX Level	5.063906	4.337625
GAF- MLP- First Diff	2.310392	2.552624
GAF-CNN- Fractional Diff	2.683462	3.069464

Comparing the GAF-CNN models, we observe that smoothing the data and reducing noise can improve prediction performance. As Livieris et al. mentioned, while deep learning has shown efficiency in forecasting time-series prices, it has also been shown to produce unreliable forecasts. The reason is some properties of time series, such as noise, high volatility, and, most importantly, the lack of stationarity.

Step 4

The results in the table indicate the following:

- MLP models outperform the GAF-based models.
- Making time series Stationary and smooth improve the result.
- The Log-Diff model achieves the lowest test error (0.697), so that first differencing improves predictive performance.
- The Frac-Diff models exhibit overfitting, with low training MAE but very high test MAE.

- The GAF-CNN models show higher errors, the reasons can be the small dataset (2,700 observations) and the short 15-day input windows, which limit the CNN's ability to learn long-term dependencies in the VIX series which have mean reverting behavior
- The GAF-MLP – First Diff model performs better than the GAF-CNN models, suggesting that connected networks can extract useful features from GAF images more effectively than CNNs when data is limited.

Overall, these findings shows that for small datasets, simple MLP architectures with differencing can outperform more complex GAF-based CNN models.

Model	Train Error (MAE)	Test Error (MAE)
MLP – Levels Model	1.043033	1.063087
MLP – Log-Diff Model	0.963735	0.697469
MLP – Frac-Diff Model	1.054819	7.581117
GAF-CNN – VIX Level	5.063906	4.337625
GAF-MLP – First Diff	2.310392	2.552624
GAF-CNN – Fractional Diff	2.683462	3.069464

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